One fungus, many hosts? Symbiosis between *Daldinia loculata* and its many plant hosts

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Abstract

The endophytic species, *D. loculata* is found in diverse photosynthetic hosts. In my research, I address the question of whether *D. loculata* is a generalist and able to associate with this apparently broad host range, or is there cryptic specialization for each host? We tested to see if various strains of *D. loculata*, isolated from different hosts, are able to grow in association with moss, and if so, how interactions between *D. loculata* and moss affect moss growth. Our work thus far has provided a foundation for future studies regarding host association at the cellular and functional level.

What is a fungal endophyte?

- Class of fungi that live inside of plants but unlike known pathogens do not cause obvious signs of disease
- Found living in diverse plants all over the world
- Some may increase plant resistance against herbivorous animals and insects in some plant hosts: also known to confer other fitness benefits in response to both biotic and abiotic influences(1,2,3)

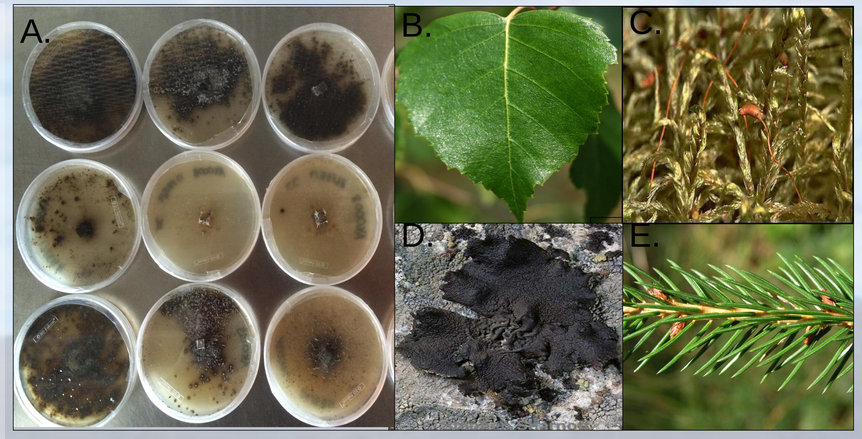


Figure 1. Exceptionally wide host range of *D. loculata*. (A) Cultures of *D. loculata* strains isolated from (B) *Betula nana* leaf litter, (C) *Pleurozium schreberii* , (D) lichen, and (E) *Picea glauca*.

Q1. Will individual isolates from different plant hosts colonize moss equally well?

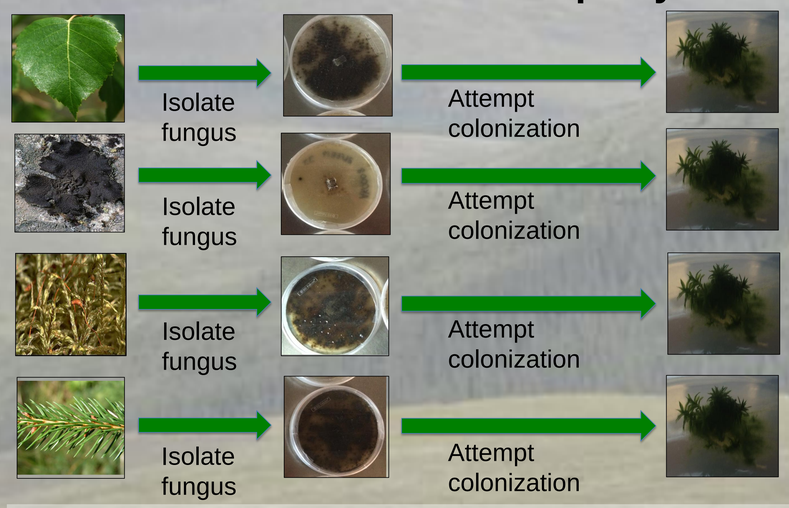


Figure 2. Strains of *D. loculata* from diverse hosts and attempt to colonize a common host, moss. The left most column depicts the hosts of origin. We will inoculate laboratory grown moss with the different fungal strains. The moss was grown on Prague's media and subjected to a 12/12 light cycle; the temperature was held at approximately 25°C.

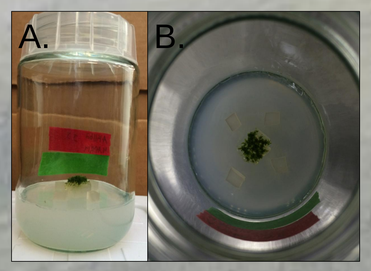


Figure 3. Host colonization experimental setup. (A) Modified baby food jar with Prague's media. (B) Moss is surrounded by four pieces of agar containing mycelium.

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Q2. Does *D. loculata* respond to the presence of the moss?

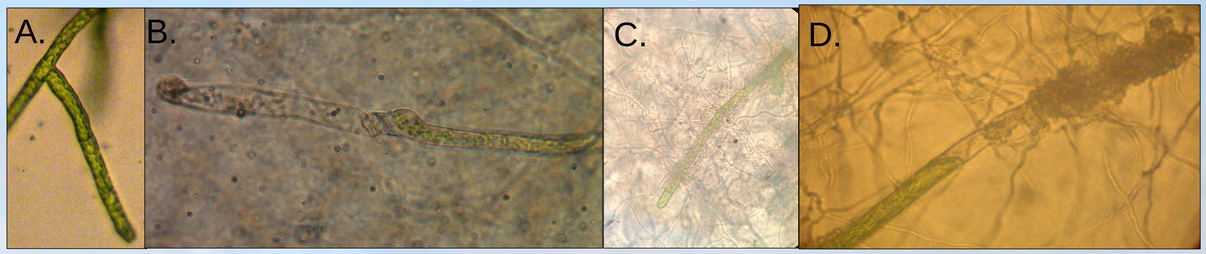


Figure 4. Microscopic images of protonema and *D. loculata*. (A) A normal protonema not in the presence of *D. loculata*. (B) An abnormal protonema in the presence of *D. loculata.* No physical contact is visible between the fungus and the protonema, suggesting diffusible substances in the environment are effecting the protonema negatively. (C,D) Physical interaction between protonema and fungus.

Q3. In co-culture, do different strains affect moss growth differently?

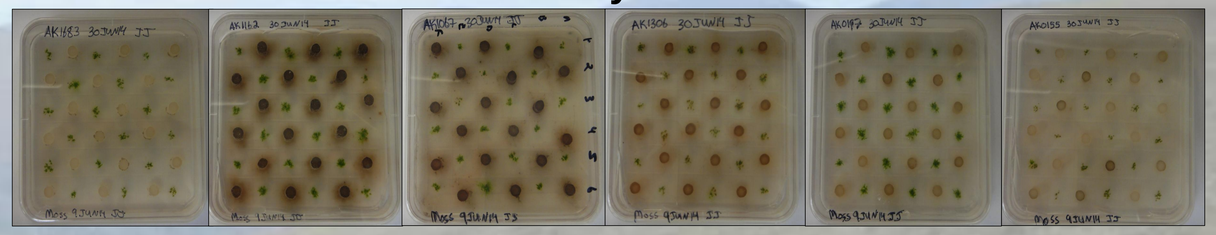


Figure 5. Six different strains of *D. loculata* were used and each affected the moss in different ways. Moss colonies were measured using a ruler. The color, health, and vertical growth of the moss were determined visually. An inverted microscope was used to make observations at the cellular level.

Results

Q1: Will individual isolates from different plant hosts colonize moss equally well?

- Staining and microscopy did not demonstrate colonization of moss by *D. loculata.*
- Surface sterilization of the moss and culturing also did not demonstrate the presence of endophytic fungi within tissues of the moss.
- Our current method of host colonization may not be effective and a new method of colonization will have to be developed.

Q2: Does *D. loculata* respond to the presence of the moss?

- Abnormal physical changes such as loss of chloroplast and bulbous growths of protonema occurred in the presence of *D. loculata.*
- Physical interactions between the fungus and protonema were (see Figure 4D) not common.

Q3: In co-culture, do different strains affect moss growth differently?

- Different strains vary in their effects on moss in co-culture.
- The moss colony size, color, and vertical growth were affected differently by different *D. loculata* strains.

Conclusion

We observe variation in the effects of individual strains on moss growth, and current results do not demonstrate endophytic colonization of moss. Our tentative conclusion is that *D. loculata* may not be a generalist symbiont of these diverse plant hosts.